

Designing Interactive Technology Roadmaps: A Visual Analytics Approach

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Abstract. Technology Roadmaps (TRMs) have become indispensable tools for aligning research and development efforts with strategic objectives. Highly graphical in nature, TRMs often receive criticism for their static nature and designs. In this study, we propose the integration of visual analytics (VA) techniques to enhance the usability, insights, and interactivity of TRM applications. Following a user-centered design approach, we collected requirements from seven stakeholders and developed two interactive prototypes: a flat-style dashboard design and a nested-style drill-down design. Comparative user evaluations revealed differing strengths favoring dashboard design particularly for its simplicity and ease of access. Based on our findings, we present a set of generalized design guidelines and introduce a Three-Stage Design Model that guides developers from structuring data to enabling user sensemaking in TRMs.

Keywords: Technology Roadmaps · Visual Analytics · Design Study.

1 Introduction

In an era of rapid technological advancement and increasing uncertainty, technology forecasting tools have become indispensable for helping organizations and industries navigate future challenges. These tools play a critical role by predicting trends, assessing technological performance, and understanding their impacts on industries and markets. One widely used instrument for technology forecasting is the technology roadmap (TRMs), which visually represents the path of technology development and adoption over time. Technology roadmapping is defined as a process that combines structured systems thinking, visual methodologies, and participatory approaches to address organizational challenges, support strategic planning, and foster innovation management at different levels [22].

Highly visual in nature, roadmaps facilitate communication across diverse stakeholder groups to enhance decision-making and strategic alignment [14]. However, studies ([4], [17],[16]) have repeatedly criticized the graphical designs of TRMs, highlighting their poor development and lack of intuitive design principles. This has resulted in roadmaps often being ignored due to limited reliance [33] by practitioners and researchers alike. Kerr and Phal [16] emphasized that roadmapping should be approached as a visualization process for communication rather than a static product, requiring thoughtful integration of design principles, human perception, and visual metaphors.

Despite them being highly infographic, most technology roadmaps remain largely static, with limited interactive and visual analytics capabilities. Some of the design challenges addressed are information overload, off-putting color schemes, and distracting visual clutter [14]. Further, it includes the difficulty of conveying complex, multi-layered information [33] in an intuitive format and the inability to integrate real-time changes or dynamic interactions between evolving technologies. Such shortcomings can be crucial for fast-changing automotive industry, which is experiencing a shift due to current developments in electrification, connectivity and automation. Companies are therefore faced with the question of which technologies and products will be relevant in the future.

For project [hidden blind review], a Technology Roadmap [hidden blind review] is created as a website to visualize the expected technological developments of passenger cars until 2035 on the basis of more than 40 modules, with around 150 technologies in the areas of propulsion, connectivity, and automation. This roadmap outlines four parallel development paths: battery-electric drives, hydrogen-based drives, automated driving and synthetically generated fuels. Although, this roadmap provides foundational guidance to their main stakeholders: small and medium-sized enterprises (SMEs) and research institutes, its static nature presents limitations in user engagement, data exploration, and decision-making support. Further, it fails to capture deviations from planned trajectories or visualize the interdependence of emerging technologies which is also shown in a Technology Roadmap in Appendix XX.

Incorporating Visual Analytics (VA) dashboards and storytelling visualizations can significantly improve the platform by intuitively presenting the complex information and making the platform interactive, insightful, and user-friendly. To attain this, we followed a participatory design approach [11], starting with the identification of user needs, followed by the development of functional prototypes, and culminating in iterative evaluation sessions with domain experts. The insights gained from this process informed the derivation of a set of design guidelines and the formulation of a Three-Stage Visual Analytics Design Model tailored to TRM development. While the guidelines emerged organically through our design, evaluation process and our literature analysis, the resulting framework addresses a critical gap between the fields of TRM and visual analytics. Our investigation revealed that most existing TRMs are static and infographic-based, lacking the dynamic exploration capabilities that VA systems offer. Moreover, much of the prior work ([23], [16], [33]) has primarily focused on what to visualize and provide limited guidance for implementing interactive, user-centered VA tools in this domain.

1.1 Contribution

Our work represents one of the few VA-driven, user-centered design studies focused specifically on TRMs. The results are grounded in both empirical evidence from users in the vehicle technology sector and broader design knowledge from the VA and Human Computer Interaction (HCI) literature. The derived design guidelines are intended to be domain-agnostic, supporting the development

of TRM systems across diverse fields. Additionally, we propose a Three-Stage Design Model that outlines a structured and practical pathway for developers seeking to build interactive, stakeholder-friendly TRMs. This framework helps bridge the gap between roadmap content planning and effective visual interface design, offering a foundation for future research and practice in VA-enabled technology planning tools.

In this paper, we first present the insights gathered from our requirement analysis study (section 2), to identify user needs and expectations for a TRM based VA application for vehicle technologies. Next, we used these insights into the development of two design prototypes and presented it in section 3. Further in section 4, we present the result of evaluation done on the developed prototypes, focusing on its usability and effectiveness. In section 5 and section 6, we further contribute by presenting the derived guidelines and a three-stage design model for VA in technology roadmapping tool. Finally we conclude and summarize this paper in section 7.

2 Requirement Analysis Study

The goal of our requirement analysis study was to gather requirements from different user groups about their current understanding, usage, needs, and expectations from the visualization tool representing TRMs. Participants were first shown an introductory presentation on TRMs. Then questions were asked about their familiarity with TRMs, their reasons for using them, and their specific information and visual preferences. The questionnaire is available in the Appendix X. It was developed by taking assistance from the professionals in vehicle technology management, HCI designer and a visualization scientist. The requirement analysis study was conducted with three main stakeholder groups, who either provide the data or use the result of this roadmap. There were 3 academics, 3 industry experts, and 1 participant from state administration. The study was conducted online via a face to face interview and a questionnaire to gather their feedback. It lasts for an average of 20 to 30 minutes for each participant.

2.1 Result

In the following we have summarized the result of our requirement analysis study.

- **Familiarity with TRMs:** On the 7-point Likert score, participants were asked about: their familiarity with TRMs, how often they use them, reasons for using them, and their specific information preferences. We found out a high level of awareness and familiarity about TRM tools among users with a mean average score of 5.14 out of 7. However, results also show that participants were not frequently using them (3.14 out of 7). This discrepancy among awareness and usage indicate a critical usability gap. Thus, making it

more significant that TRM tools need to be both informative and accessible with limited learning curve.

- **TRM objectives:** We derived three key motivations for using the technology roadmap tools: 1) gaining insights into technological evolution and interdependencies; 2) facilitating communication and collaboration among stakeholders and 3) identifying trends to guide strategic decisions and product development.
- **Age-based user preferences:** We see differences in user preferences across different age groups. Users within age group 25-34 prefer complex, dynamic, and advanced visualizations (e.g., sunburst and network diagrams). They have particular interest in knowing the temporal trends and spans, relationships and different types of technology fields. The one within age group 35-44 value structured, focused visuals like Gantt Charts and narrations with filtering options. User within age group 55-64 prefer known and straightforward visualizations such as tree diagrams, reflecting their need for clarity and minimal complexity. Moreover, they need highly accessible tools with automatic updates, translation options and detail information.
- **Sector based preferences:** Academics primarily utilize TRMs to gain insight into current developmental progress and to assess project milestones and technological readiness level (TRL). Industries leverage TRMs to stay updated on evolving technological trends, driving innovation and maintaining competitiveness. State Authorities utilize TRMs for comprehensive market research and analysis in technology-related decision-making processes.
- **Other prolific insights:** Timelines and timespan are the most important artifacts that are needed to be shown in these roadmaps. However, most participants were less interested in mega trends but were more interested to know the small trends in the evolution that can guide them to timely opt for corrective and transformative measures. It should be noted here that majority of visualisation studies [34] use deviation or big changes in the trends as a good contributor to highlight time of interests in the charts. However, it seems untrue for TRMs use case. As temporal trends were the important insight for them, thus gantt chart was selected as a prominent visualization type, followed by sunburst diagram and other network diagrams. Furthermore, one of the important information to show on the TRMS for vehicle technologies is, 'Market Share' followed by 'Patents' and 'Key Companies'. The most needed option for the tool is to have interactive features, automatic updates, and variant filtering options. Participants are further interested in detailed information with external links to data or information sources, translation, or narration options. These features thus needs to be highly customizable, so that it can be easily adopted to diverse demands and can be easily configurable. These requirements, strengthen our claims of using interactive narrative-based visual analytics techniques like storytelling visualizations to present the technology roadmap for its effective and broad usage.

Initial findings from this study, underscore the importance of tailoring the TRM application to meet the diverse needs of users. Table 1 shows the key derived requirements from this study. Lessons learned from these interviews were applied to the design and development phases, to ensure that the tool seamlessly aligns with user expectations and enhances their overall experience.

Table 1. Design Requirements for Technology Roadmapping tool

Components	Requirements
Data and information artifacts	Timelines and timespans, relation between different technology modules and fields, market shares and patents.
Visualisation artifacts	Gantt Chart, Sunburst and Network Diagrams.
Interactive Features	Filtering options, onclick narrations, Hovering, popups
Other features	Automatic updates and detailed information (documentation)

3 Technology Roadmap Design

The goal of TRMs is to translate complex system-level strategies into representations that are easily comprehensible across diverse user groups. Rather than merely displaying information, such visual interfaces enable users to interpret trends, assess uncertainties, and explore potential future developments based on the data relationships and pathways made visible. To make these complex visual representations truly accessible and actionable, narrative and storytelling techniques could play a critical role. Storytelling in visual analytics goes beyond aesthetics or layout. It provides structure, context, and a cognitive path through the data [29]. In the context of TRMs, by guiding users through sequential, contextual, or thematic progressions, Storytelling along with narrative designs help bridge the gap between raw technical data and actionable insight [18,10]. In our study, we operationalize two distinct narrative strategies from the findings of [29]: a *Dashboard-based design*, which offers a flattened overview with structured entry points, and a nested style *Drill-down design*, which allows users to progressively unfold information through guided transitions. Further, based on the Table 1, we decided which visualisations would be used for representing what relationship types and what information was mapped. These prototypes were developed with Oracle APEX environment and TKBW database on the evolution of technologies over time. Oracle APEX enables rapid creation of scalable and secure data-driven applications through a powerful low-code development platform. The database use TRL and Manufacturing Readiness Level (MRL) for modules from 2019 to 2035 with data based on expert responses to the technology profiles.

3.1 Dashboard Design

The **Dashboard** design prototype first provides an overview of the essential components, connected and comprised on the same page. Then through interactions, user delve deeper into specific aspects of each components. The Dashboard design is shown in [Figure 1](#) and its workflow in presented in [Figure 2](#). There are three primary visualizations through which one can explore data and dive deeper into other options: 1) Sunburst Diagram shows hierarchical data structure based on vehicle types, modules, and technologies. Clicking on each section opens up further details, 2) The network diagram focuses on highlighting connected elements - competing technologies, keywords and competencies and 3) The Gantt chart emphasizes the chronological representation of project timelines and time spans. The Gantt chart is combined with box plots to provide better view of the time span. Checklist and search option is provided for the user to further add or remove the needed technologies, module and competency types. As this tool was developed for German users thus the main language is German, however, on clicking on button "Sprache auswählen", user can change the language to English. The workflow shown in [Figure 2](#) also guides about the usability of this prototype. Further a video is provided in Appendix x to see this prototype in action.

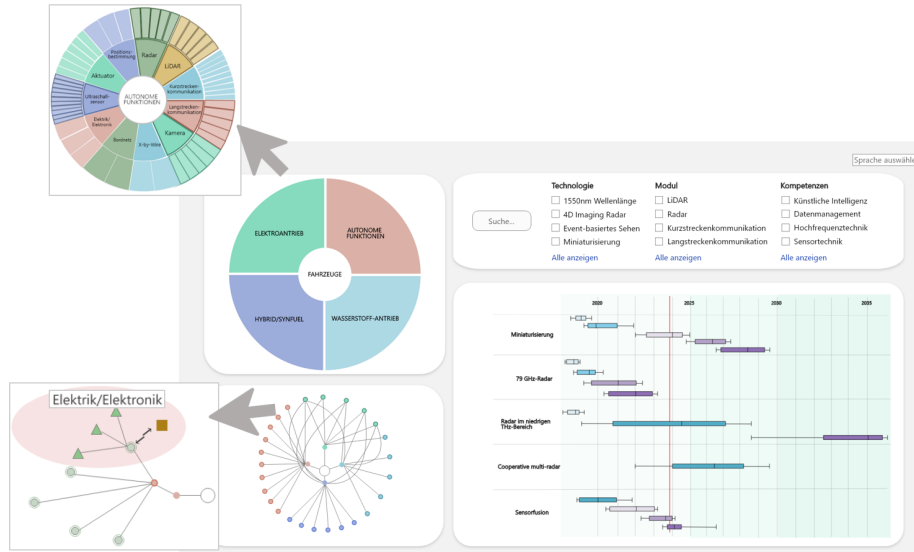


Fig. 1. Dashboard Prototypes. Screenshots of clicks are embedded for better understandability of the prototype.

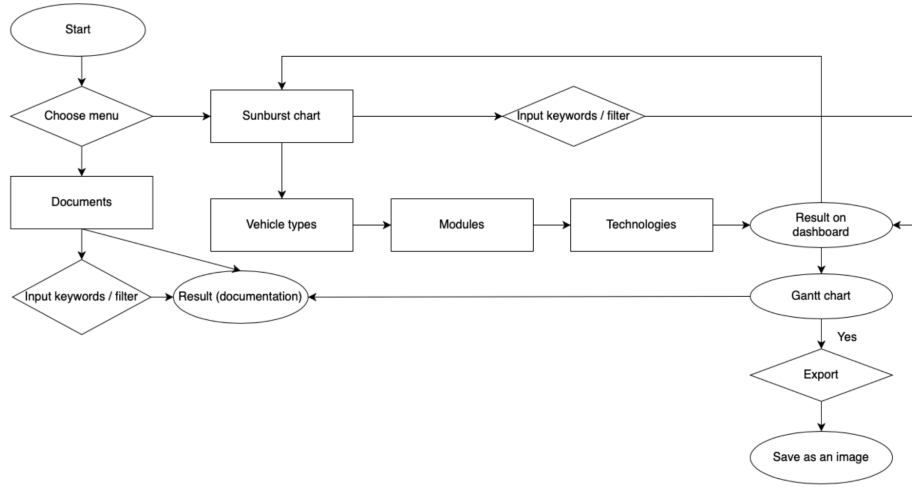


Fig. 2. Flowchart depicting the interaction flow in our Dashboard design prototype.

3.2 Drill-Down Design

The **Drill-down** design prototype first presents general broad themes and then allows the user to select specific themes to reveal additional details and backstories. The Drill-down design is shown in [Figure 3](#) and its workflow is shown in [Figure 4](#). It tries to provide a high level view of the data through one Network diagram which is nested into four levels. The upper most level shows the vehicle types and how they are connected to each other through different edges. The other nested levels show modules -> technologies and other details. These levels are also shown on the navigation bar above the network diagram. Clicking on the nodes on this diagram open up modules associated to the clicked vehicle types. Clicking on each module, shows the timeline and timespan of the technologies and other details in a Circular Gantt Chart. This layout unfolds the option based on the semantic layering of different vehicle types -> Modules -> Technologies associated with it. Each version of this diagram comes with its own interactive components and insights. The workflow shown in [Figure 4](#) also guides on the usability of this prototype. Further a video is provided in Appendix to see this prototype in action.

4 Evaluation

Further we conducted a usability evaluation of our two developed prototype designs. Participants were asked to perform real-world scenario-based tasks to assess the effectiveness of each design. The objective of this study was twofold: first, to introduce users to our TRM designs tailored to their feedback and requirements, and second, to systematically compare the two interface designs to

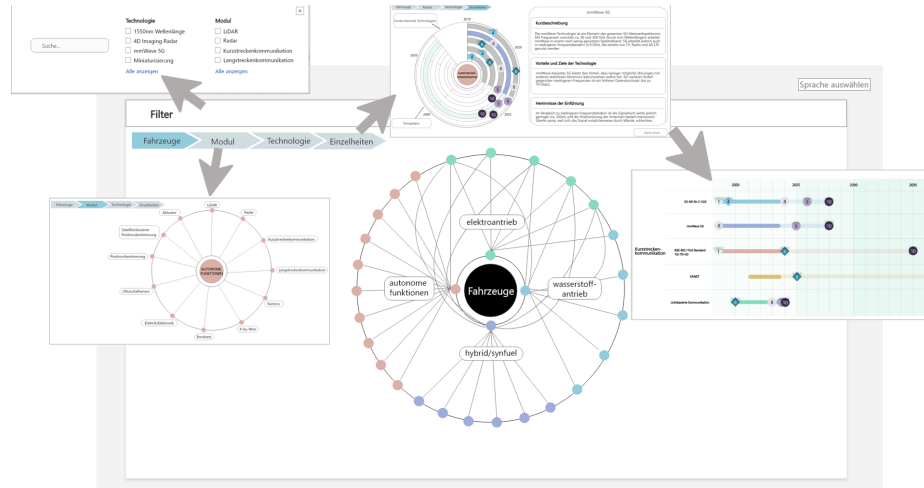


Fig. 3. Flowchart depicting the interaction flow in our Drill-down prototype.

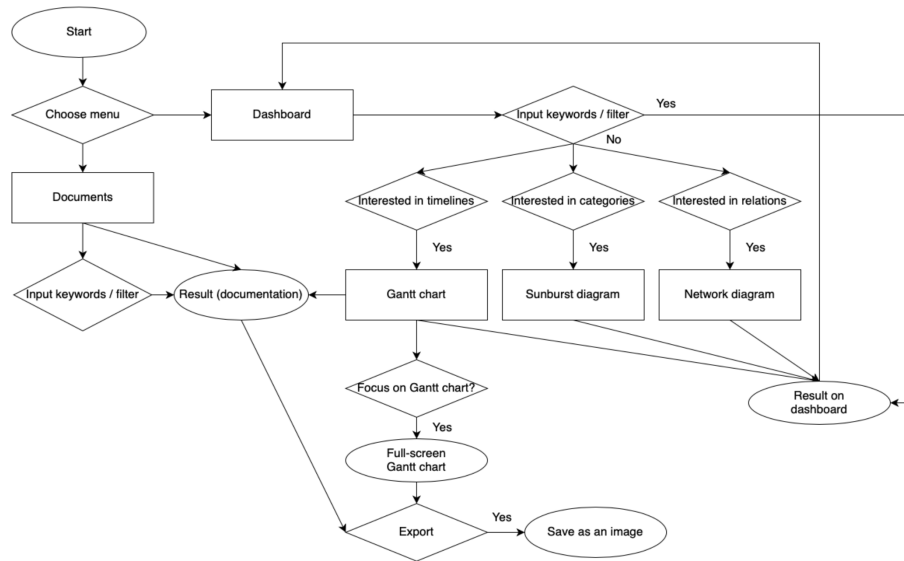


Fig. 4. Flowchart depicting the interaction flow in our Drill-down prototype.

determine which one better aligns with the needs of our stakeholders. The feedback was collected via a questionnaire which was developed by taking assistance of QUIS [6] and NASA TLX [9] survey templates. The study involved 10 participants. The questionnaire is available in the Appendix XX. The results are summarised below:

1. **Task performance:** Participants were asked to performed tasks which were then evaluated to measure performance. Performance is measured in two ways - errors and response time. Errors refer to the incorrect answers of the participant's responses from the performed tasks [25]. Response time is the time it took a participant to complete a task. The performance with Dashboard design is better as it has an average error score of 7.3% and response time of 56.43 seconds. The average error score for the Drill-down is 12.9% with the response time of 78.36 seconds.
2. **Design choices:** Participants were asked questions about their design choices related to three different pair of alternatives. First comparison was related to the layout of the prototypes. Between Dashboard design and Drill-Down design, majority votes (60%) were gone to the Dashboard design. The reason provided by participants were - simple and straightforward. Second comparison was between One Liner Gantt Chart and Gantt Chart with Boxplot. Here participants preferred One liner Gantt chart (70% votes) primarily because they were already familiar to them. Third comparison was between Sunburst Diagram connected with Network diagram and Network Diagram alone. Here, 80% of our participants opted for the connected Sunburst Diagram, owing to the clear overview of the data and insights thereof from this diagram.
3. **User experience:** We measured user experience across four main categories: Usability, Accessibility, Learning Curve, and UX/UI elements. Each category included specific subcategories to capture detailed user feedback and assess different aspects of the experience. These were then rated using a 7-point Likert scale.
 - (a) Usability: We analyze overall usability of these designs based on four metrics: Ease of use, Satisfaction level, Stimulating and Flexibility. The results are shown in Figure 5. Dashboard scored higher than Drill Down for *Ease of use* and *Satisfaction*. Whereas in *Engagement* and *Flexibility*, scores of drill down is slightly better than Dashboard.
 - (b) Learning curve: The parameters on which Learning Curve was scored were: Informative, Easy to learn and start, Straightforward, Exertion Level, Stress Level and Experience Level. Dashboard again scored higher particularly it being *Informative*, *Straightforward* and less *Stress Level* and *Exertion Level*. Dashboards has a slight advantage in performing task straightforward due to its flat layout. Due to the nested layout of Drill-down, the amount of effort needed to perform the task is more and thus the Learning Curve for Drill-Down is higher.
 - (c) Accessibility: We evaluated different accessibility parameters of our designs: orientation on the screen, interaction level, use of reference material, information density and information organisation. Dashboard is

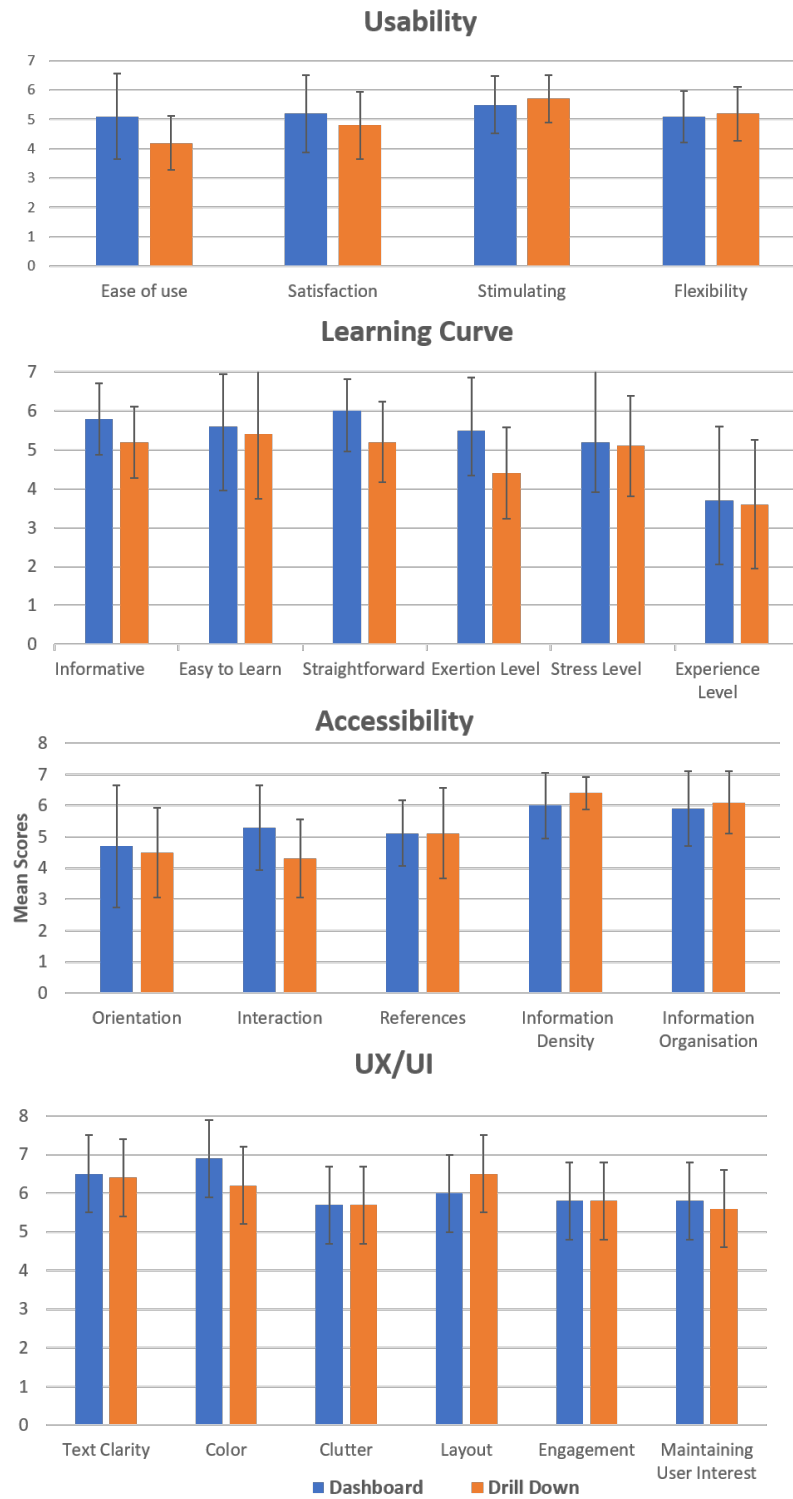


Fig. 5. This figure shows the user evaluation results comparing the Dashboard and Drill-down design prototypes. The bar chart presents mean scores with error bars showing deviation across four key categories: Usability, Learning Curve, Accessibility, and UX/UI.

rated higher across most *Orientation*, *Interaction*, and *References*. Drill-Down strongly outperformed dashboard in *Information Density* and *Information Organisation*.

- (d) UX/UI: We evaluated User Experience and User Interface (UX/UI) via following parameters: text clarity, colorscheme used, cluttered interface, attractive layout, user engagement and maintaining user interest. Dashboard scored higher on most categories i.e *Text Clarity*, *Color*, *Engagement* and *Maintaining User Interest*. Both prototypes have been equally scored for cluttered, however, Drill-Down has scored better for its Layout design.

The results are visualized in Figure 5. Across all evaluation dimensions (22), participants ($n = 10$) rated the Dashboard design (blue) higher in terms of overall usability, learnability, and accessibility, the Drill Down design (orange) was rated more stimulating and better in information organization. Results shows a direct preference towards Dashboard type flat designs. Drill down has also resulted good more well organized and information dense design. As majority of our participants were within the age group of 25-35, this may be concluded that young age group prefer more direct information and exploration.

Further we have tried to understand the dependencies and correlation among individual design factor. Here our goal is to explore which factor relationships are consistent, and which are design-dependent. For that, we used Spearman correlation Test [1] for the discrete datasets.

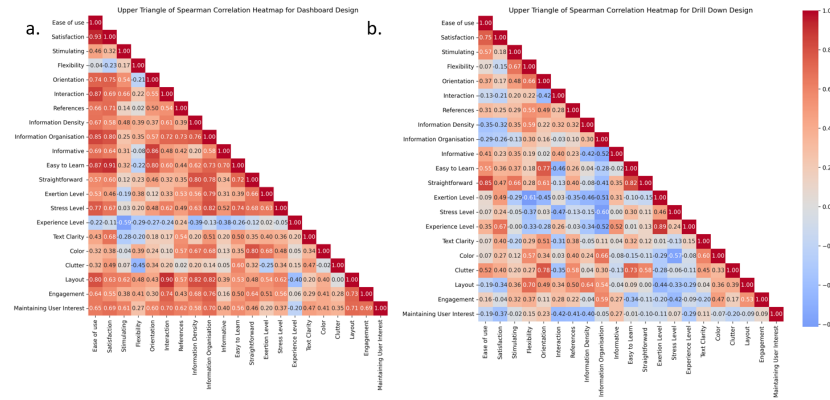


Fig. 6. This figure shows the upper triangle of spearmen's correlation matrix for correlation among different factor. Red color shows the level of positive correlation and blue color shows the negative correlation.

As per Spearman Correlation Interpretation Thresholds [1], above 80% is very strong, above 60% is strong and then below zero is negative. Figure 6a shows that Dashboard design has most positive correlation. Those users who rated this

design prototype as easy to use (*Ease of use*) has also rated it high in *Satisfaction*, *Interaction*, *Information Organization*, *Easy to Learn* and low *Stress level* and Good *Layout*. There are very few and mild negative correlations. However Figure 6b shows most of the negative correlation specially with *Experience Level*, *Information Organization*, *Stress*. This behavior suggest increased tension between design elements and varied user reactions, particularly among experienced users.

By analyzing the overall results, we identified both stable and volatile UX factor relationships. While core links like *Easy to Learn* $\leftrightarrow$ *Satisfaction* remained strong across designs, other relationships such as *Layout* $\leftrightarrow$ *Engagement* and *Information Organization* $\leftrightarrow$ *Ease to Learn* proved sensitive to design context. Our results highlights the importance of validating interface patterns across multiple conditions rather than assuming universal user behavior.

5 Design Guidelines

Based on our experience from this study and literature analysis, we have derived design guidelines for constructing an effective visual analytics application for technology roadmapping.

1. Follow design principles: General design principles laid down by scholars [20] [21] and researchers are critical and should be followed in the construction of either static infographics or interactive applications. These principles guide users in the creation of clear, uncomplicated, clutter free designs. For example as per the principle of Consistency - a visual metaphor that looks like button should behave like button and the same design should be used throughout the application to avoid confusion. Another one, as per the principle of Alignment, grouping the similar items together visually make scanning quicker and support comprehension. Findings from our study as well as previous studies [2] [1] have highlighted the role of creating clear, simple, and straightforward designs with user friendly layout which ultimately reduce unnecessary cognitive load.
2. User-centric design process: Studies have emphasized on rapidly involving stakeholders in the design and development process [15][32]. Construction of TRMs have been considered as a collective task [33] and thus it is essential that different kinds of stakeholder (data providers, decision makers, users getting influenced from those decisions etc.) should be in a loop during and after the development process.
3. Scalability and Flexibility: Scalable and Flexible designs [37][13] ensures roadmaps can be expanded to future datasets and domains. They are adaptable to new updates both in data and requirements. Views could be easily customisable that can cater the need of diverse stakeholders with their needs [1] [7]. Further, scalable and modular design could be a solution to keep TRMs alive and ongoing, a challenge identified by [23]. Modular components and tailored workflows can meet specific user demands effectively. Some of

the ways to achieve scalability and flexibility: choosing scalable databases like Graph or NoSQL types, use modular data pipelines, use more hybrid and hierarchical visualisations that are scalable and can show multiple dimensions, compatibility with different devices like mobile phones, tablet etc, allow user to hide and show views based on their requirements, provide role based access and provide more interactivity, filters and drill down options. Allow seamless integration of data from multiple resources (for example patents, market shares etc) and real time updates ensures TRM remain up to dates and relevant for the users, particularly in industries like automotive, where innovation timelines are tight.

4. Collaboration and Knowledge Sharing: Features allowing collaborative input and document sharing were noted as valuable additions to roadmapping tools as shown in our and also previous studies[1] [15]. Collaborative visual features to allow cross-departmental input and sharing with external stakeholders is seen as successful implementations in large organizations [1]. Shareable views [35], collaborative updation and commentaries and exporting the results in different formats are some features that can assist in the collaboration and knowledge sharing.
5. Guidance and Assistance: To decrease the learning curve [33] and for better insight generation, it is important to include different guiding and assistive features in the TRM applications. Here, guidance means to allow all those features that provides knowledge about the proper functioning of the tool. Such as providing proper onboarding, tutorials and contextual tooltips. Whereas assistive features are those which help to cater the need of different types of stakeholders irrespective of their age and other differences. As also showed in our study, that differences in age groups leads to different requirements for the TRMs. Thus to cater the demands of varied users, a developer can include different options such as editable colorschemes, zooming features, minimize and maximize different views, alternative views of similar visualizations, translation options etc.
6. Data Security: Addressing security concerns related to proprietary data, as mentioned in broader literature [], ensures user confidence in the tool. Moreover, user login to a secure system and providing role based views and privileges on the system could further assist in better management of data exposure to different types of interested parties.

Further options could be related to domain specific requirements, for example if data demands comparison among geographical regions than provide both local and global perspectives in technology trends, adoptions, and regulations. Another is providing specific options for scenario analysis as proposed by Son et al. [31]. These guidelines help developers to decide on various features and the scale of work needed in the development of TRMs.

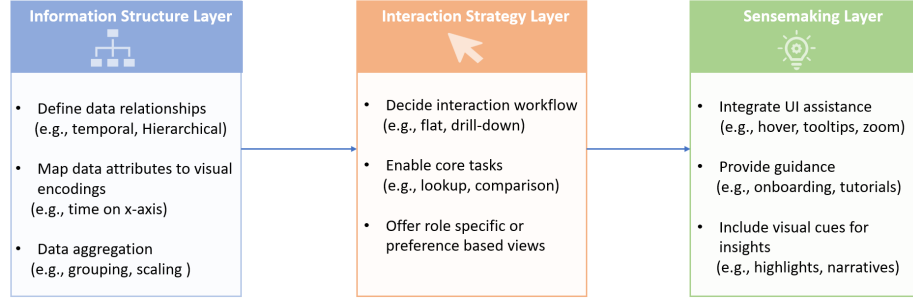


Fig. 7. Three-Stage Design Model for Visual Analytics in Technology Roadmapping. It guides TRM developers from data structuring to user interaction and then to insight generation.

6 Design Model for Visual Analytics in Technology Roadmapping

Further, we propose our Three-Stage Design Model for Visual Analytics in TRM development. This framework is derived based on our own developmental path for TRM in vehicle technology, as well as, by taking inspiration from established cognitive and interaction frameworks in information visualization and human-centered design. This model builds up from data structure to user interaction and ultimately to communicative and interpretive visualizations. However before starting to implement this model, a developer has to understand the domain and data well by involving stakeholders and reviewing relevant literature.

1. Information Structure - *What has to be visualised?*

In this foundational stage, developer model and understand the gathered data as a beginning of a design process. It corresponds to the data abstraction and encoding phase of the visualization pipeline, as conceptualized in classic models by [5] [30]. This stage can be divided into three layers:

- (a) Define data relationships: Knowing the data well, first developer has to decide the relationship that are depicted from the data elements. As these relationship further corresponds to the user goals that can further derived corresponding visualisations [12]. For example: if elements of the dataset shows series of date and time, then it shows temporal relationship and thus can be visualised by temporal visualisation (gantt chart, temporal line or bar charts). Knowing this relationships within data elements at the earlier stage assist developer in searching for suitable visualisations and also the other alternatives of the similar visualisations. Some suitable questions at this stage would be:

** What kind of data is involved? (timeline, dependencies, modules, metrics, actors)?*

** Is there any apparent data relationship or structures and if so then what are they? (is it temporal, hierarchical, networked?).*

* *How can this relationship be visualised (e.g., hierarchical -> sunburst, treemap)*

- (b) Map Data attributes to visual attributes: Once a developer knows which visualisations to use, next step is to map different data attributes to visual attributes. Developer tries to answer the questions *What data elements will be mapped with what visual elements of the chart?* For example in Figure 1 nodes in the circular network diagram are the vehicle types). For this, it is important that developer knows about the types of data attributes well. *Is it categorical, nominal, temporal (Date/Time) etc.?* For example in temporal line chart 'x' axis would be time line and 'y' axis could be market shares and color of the lines could be different market types.
- (c) Data aggregation or categorization: Visualization relies on structured aggregation for clarity. As TRMs usually spans from long time horizons (e.g., 2025 → 2050), there are multiple scales of technologies (system → subsystem → module → component) and has diverse stakeholders (policy, R&D, SME, industry). To structure this data, aggregation and categorization is an important step. Aggregation provides a semantic anchoring and without that user gets overwhelmed in multiple level of details. Here the user has to decide *What levels of aggregation will be shown? (e.g., years → months etc for temporal and different nesting levels for sunburst diagrams.)*

The result of this stage would produce a table similar to the one presented in Table 1 but with more details. For this stage a novice visualization developer can also take help from already established data and visualization taxonomies and knowledgebases. For example: taxonomy by [12], visualisation catalogue [28], book [8] and other available resources.

2. Interaction Strategy Layer: *How will it be explored?*

Once the visual content is decided, the next step is to decide on the exploratory workflow of the application. This layer addresses exploratory behavior, aligning with the interaction taxonomy defined by [36], cognitive models of exploratory search [19] and narrative flow by [29]. It encapsulates user-driven strategies for interaction flow on the interfaces such as overview-first -> navigation-> progressive filtering -> detail-on-demand -> comparison. These strategies reflect how users cognitively approach roadmap data. This stage is further divided into three sub layers:

- (a) Deciding interaction workflow: Whether it is a visual analytics based or any other UI based application, interactive workflow is highly dependent on the layout of the application. For example in our prototype we used two types of layouts i.e., flat type dashboard and nested type drill down layout. Even in static infographics, a layout decide the organization of all the elements on the canvas which further intuitively guides the user focus.
- (b) Enable core tasks: Next is to decide the core tasks performed by the user in the information seeking process. For example to look up on specific elements, selecting the date range from timeline, clicking and showing

the nested hierarchies in sunburst diagram, dragging elements, entering the information, performing comparisons etc. Where some of these tasks are very visualisation specific, others are also common for interactive applications like clicking, comparisons, dragging etc. Here the integral question that needs to be answered is *What tasks should the system support?* There are several taxonomies [3] developed both for web and visualisation application that could support in this process.

- (c) User specific views: It is common in data sensitive applications that not all informations are for all stakeholders. On the other hands, based on the specific age group or interests, some users prefer alternative views (as shown from our requirement analysis study result) or have different requirements. Further, some features are only for decision makers and other are for specific type of stakeholders. To fulfill many of these goals, TRMs architecture should facilitate providing different views, access levels, alternative options etc. to the users.

3. Sensemaking Layer: *How insights emerges?*

Sensemaking goes beyond displaying results to enabling the discovery of new insights as emphasized in exploratory data analysis literature [24] [26]. This layer enhance understanding, supports interpretation and insight generation through interface elements and different visual cues such as annotations, onboarding, flows, semantic visual cues, and progressive disclosure. Rooted in theories of sensemaking in visual analytics [27], it enables users to construct mental models and draw meaningful conclusions from complex data structures. It can further be divided into three different layers:

- (a) Integrate UI assistance: The first level of support is to enhance the basic understanding of all the elements in the application. This includes good captions, hover, tooltips, labelling etc. These on demand (via clicks) assistive elements, enhance the overall understanding of the UI elements. Only when there is a good understanding and sensemaking, there could be chance for insight generation.
- (b) Provide guidance: For providing intensive understanding of the data and visual elements, a developer can provide good documentation, onboarding steps and good quality tutorials. These guiding elements will also assist in decreasing the learning curve for using the TRMs
- (c) Include visual clue for insight generations: This is a step forward where various visual cues can be used as signifiers (as proposed by Don Norman in [21]), to divert the attention to the important information. Though this step is not compulsory however studies [] have shown such assistance enhance the process of insight generation. Example of these visual cues are: highlights to important patterns, hotspots, area of interest and time of interest. For successfully implementing these options, one needs to study the related domain well to understand which deviation, pattern could be interest to the user.

7 Conclusion

In this study we propose the use of visual analytic based narrative and story-telling applications to overcome the limitations of static technology roadmapping tools. Following a user centric design approach, we first gathered requirements from seven different types of stakeholders. Results from the requirement analysis has emerged as a crucial factor, revealing diverse preferences across demographics and professional needs. Our findings highlight the importance of incorporating advanced visualization techniques, real-time updates, and narrative features to meet the demands of a fast-evolving industry like vehicle technologies. Based on the gathered requirements we developed two prototypes: a Drill-down design and a Dashboard design prototypes. A comparative user evaluation indicated a general preference for Dashboard design, which was praised for its simplicity and accessibility, making them ideal for quick insights. Drill-down designs was rated as detail oriented and better in information organization. To further understand the relationship between usability factors, we conducted a Spearman's correlation analysis. Results for the Dashboard design showed mostly positive and harmonious relationships between user experience factors, whereas, Drill-down design revealed a higher number of negative correlations. We further derived a set of design guidelines for the development of effective TRM applications. We emphasize the importance of user-centered design, flexibility, scalability, collaboration, guidance and following establish design principles for the creation of these applications. To address the gap of limited theoretical frameworks for designing VA based TRMS, we presented a Three-Stage Design Model for Visual Analytics in TRMs development. This model guides developer from the stage of data and information structuring to facilitation of user sensemaking. This model poses reflective questions at each stage and supports decision-making based on the data, user needs, and system context. Together, our design guidelines and framework provide a structured pathway for developing visual analytics based TRM tools that are both functional and insightful.

During the course of our study, we saw a limited research directly comparing flat dashboard-style interfaces and hierarchical drill-down interface designs directly in visualization studies. Although both styles are widely used in interface and web application development, their relative advantages for different types of data, users and context remain underexplored. Our literature review revealed that most comparative studies focus on different visualization types (bar chart, scatterplots), or on evaluating usability of interactive visualisations. Other studies concentrated on perfecting either of these designs in isolation. Our work addresses this gap by empirically comparing both layouts in the same context. Despite a limited sample size (7 participants for first and 10 participants for second study) our results though not significant but suggestive, indicate that interface layout choices has a measurable influence on user engagement, usability perception, and insight generation. This highlights the need of future research to systematically evaluate the usability, effectiveness and decision making impact of different VA based applications across different application domains.

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